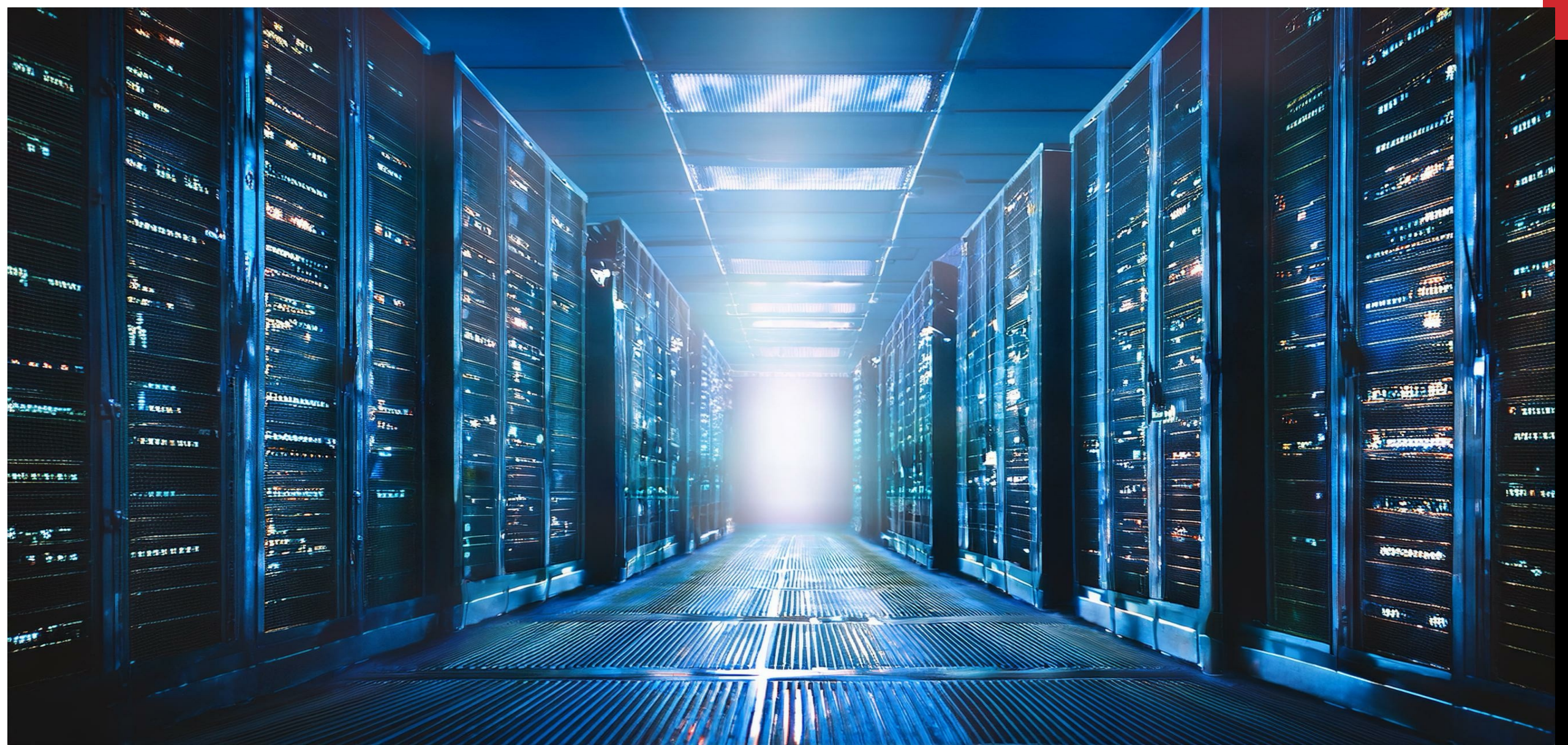


# NETWORKING FUNDAMENTALS







# COURSE STRUCTURE

**Day 1: Networking Fundamentals I**

**Day 2:** Networking Fundamentals II

**Day 3:** Networking Fundamentals II

**Day 4:** Network Security in Depth

**Day 5:** Routing Fundamentals & Capstone



# AGENDA FOR DAY 1

1. **Introduction to IT Networking:** Definition and components.
2. **OSI Model, TCP/IP, Subnetting, Essentials**
3. **CLIs & Cisco Packet Tracer**





# INTRODUCTION TO IT NETWORKING



# WHAT IS A NETWORK?

- A system that lets devices **exchange data** and **share resources**
- Composed of **endpoints** (hosts), **interconnects** (links/media), and **intermediaries** (switches/routers)
- Scales from a **home LAN** to the **global Internet**



shutterstock.com · 2520489775

# WHY DO WE NEED NETWORKS?

- **Resource sharing:** printers, storage, apps, cloud
- **Communication:** email, web, APIs, voice/video **intermediaries** (switches/routers)
- **Scalability & resilience:** many paths, fault isolation
- **Security domains:** segment users/services



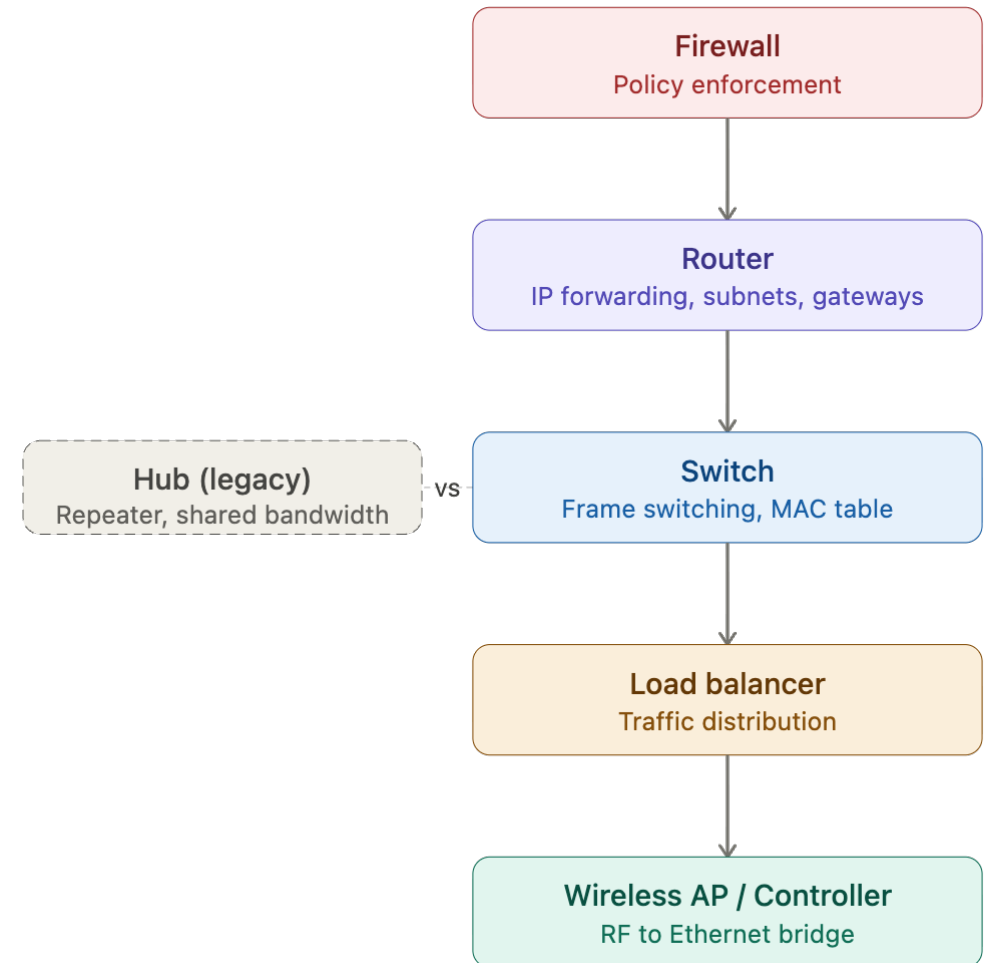
# CORE NETWORK COMPONENTS : ENDPOINTS

- **Hosts:** laptops, servers, phones, IoT
- **NICs:** MAC address, speed/duplex, drivers
- **Virtual endpoints:** VMs, containers, pods



# CORE NETWORK COMPONENTS : INTERMEDIARIES

- **Hubs** (legacy, repeaters) vs **Switches** (frame switching, MAC table)
- **Routers** (IP forwarding, subnets, gateways)
- **Wireless APs/Controllers** (RF to Ethernet bridge)
- **Firewalls** (policy enforcement)
- **Load balancers** (traffic distribution)





# CABLING & PHYSICAL MEDIA

- **Copper:** UTP Cat5e/6/6A (up to 1/2.5/5/10Gb, distance limits)
- **Fiber:** MMF (OM3/OM4), SMF; transceivers (SFP/SFP+/QSFP)
- **Optics & connectors:** LC, SC; wavelength basics
- **PoE:** power over Ethernet for APs, phones, cams

## Copper (UTP)

Cat5e → 1 Gb · Cat6 → 2.5/10 Gb · Cat6A → 10 Gb  
100 m distance limit

## Fiber

MMF: OM3/OM4 · short range · SMF: long range  
Transceivers: SFP / SFP+ / QSFP

## Optics & connectors

LC · SC connector types  
Wavelengths: 850 nm · 1310 nm · 1550 nm

## PoE

Power over Ethernet · APs · phones · cameras  
802.3af / 802.3at / 802.3bt

# WIRELESS BASICS

- **Wi-Fi (802.11)** bands/channels; APs vs controllers
- **Cellular (4G/5G)** as WAN access
- Trade-offs: mobility vs interference/coverage/planning

## Wi-Fi (802.11)

2.4 GHz · 5 GHz · 6 GHz (Wi-Fi 6E)  
Standalone APs vs controller-managed

## Cellular WAN (4G / 5G)

WAN uplinks · remote sites · failover  
Mobile users · no fixed infrastructure

## Trade-offs

Mobility vs interference · coverage gaps  
2.4 GHz range vs 5 GHz throughput

# COMMON CONNECTION TYPES

- **LAN (on-prem):** Ethernet (copper/fiber), Wi-Fi
- **WAN (between sites):** MPLS, leased line, Internet + VPN/SD-WAN
- **Data center fabrics:** L2/L3, Spine-Leaf for ECMP
- **Out-of-band (OOB):** management networks/serial

## LAN (on-premises)

Ethernet: copper (UTP) · fiber  
Wi-Fi (802.11) · private · low latency

## WAN (between sites)

MPLS · leased lines  
Internet + VPN · SD-WAN

## Data center fabric

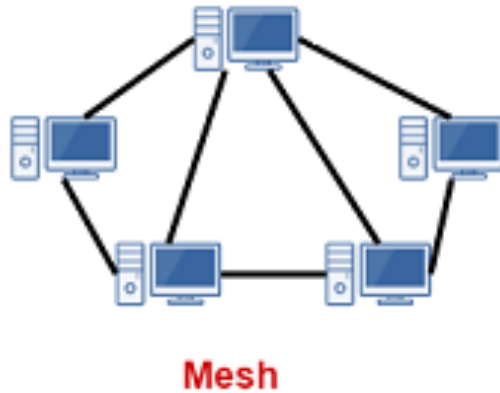
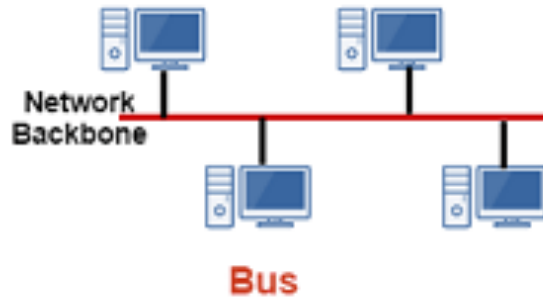
L2 / L3 switching  
Spine-leaf topology · ECMP

## Out-of-band (OOB)

Dedicated management network  
Serial console · access during outages



# TOPOLOGIES & HOW THINGS CONNECT



# KEY LINK CHARACTERISTICS

- Bandwidth vs throughput (theoretical vs achieved)
- Latency, jitter, loss—impact on apps/voice/video
- Duplex and errors (CRC, drops, discards)

# IDENTITY ON THE NETWORK

- MAC address (L2 identity on the local segment)
- IP address (L3 identity/routing between networks)
- Default gateway bridges local → remote networks

# SECURITY & SEGMENTATION AT A GLANCE

- Broadcast domains and why we segment (VLANs)
- Routing boundaries and policy points (ACLs, firewalls)
- Principle of least privilege between segments



# FROM PHYSICAL TO LOGICAL

- Physical: cables, optics, patching, racks
- Logical: VLANs, SVIs, subnets, VRFs
- Documentation links the two: diagrams, IP/VLAN tables

# POP QUIZ:

What are the components of a network ?

- A. Endpoints
- B. Interstations
- C. Intermediaries
- D. Hosts



# POP QUIZ:

What are the components of a network ?

- A. **Endpoints**
- B. Interstations
- C. **Intermediaries**
- D. Hosts



# OSI MODEL , TCP-IP, SUBNETTING, ESSENTIAL NETWORKING CLI TOOLS

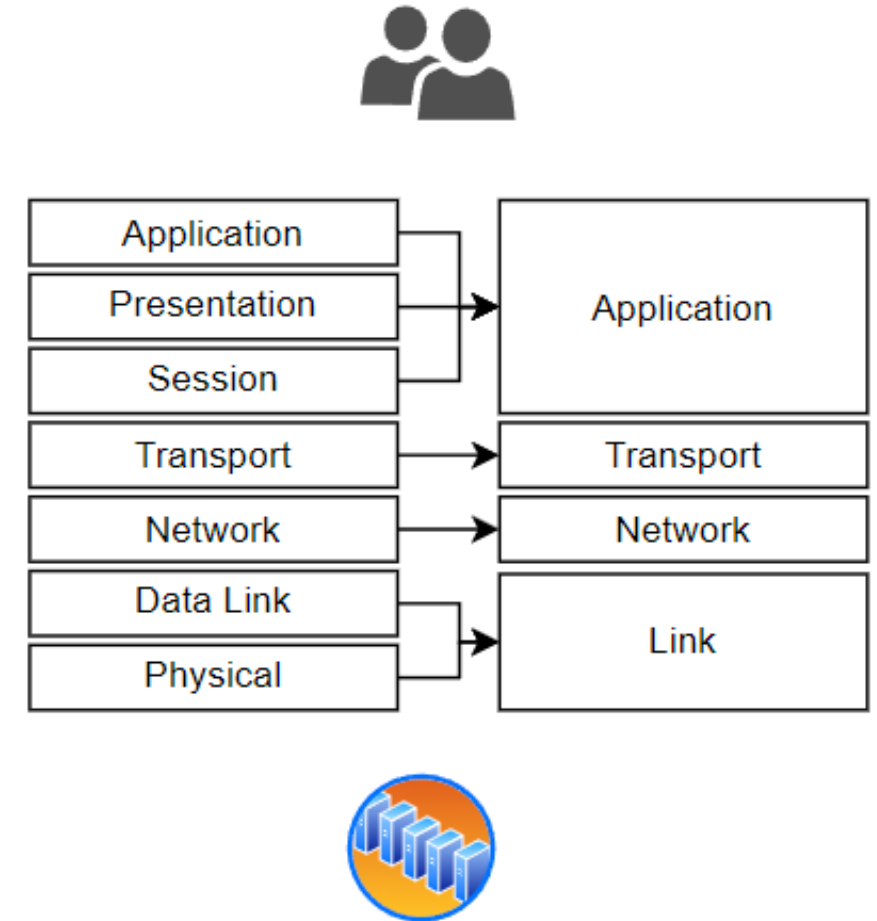
# LEARNING OBJECTIVES

- Explain OSI vs TCP/IP and map real protocols to layers
- Trace a packet end-to-end (ARP → DNS → TCP/HTTP)
- Read key fields in Wireshark (Ethernet, IP, TCP/UDP, DNS)
- Navigate IOS/EOS CLIs, use show, configure interfaces, and save



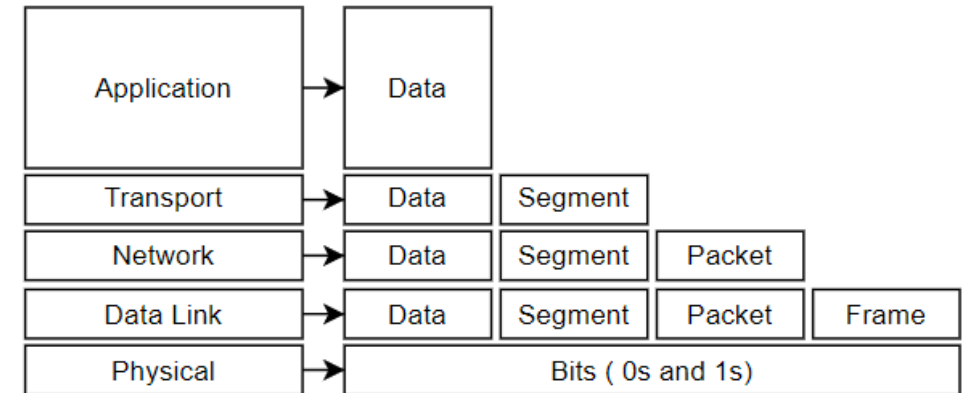
# OSI VS TCP/IP MODELS

- OSI Model (7 Layers): Application, Presentation, Session, Transport, Network, Data Link, Physical
- TCP/IP Model (4 Layers): Application, Transport, Internet, Network Access
- Relationship: OSI is a conceptual reference; TCP/IP is practical implementation.



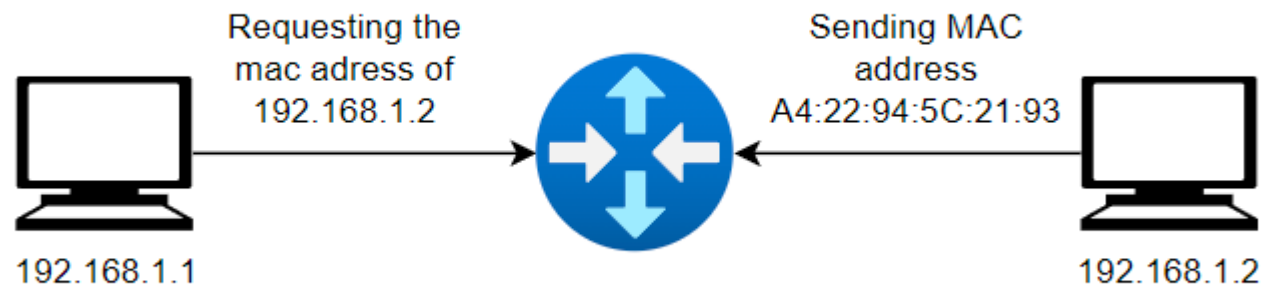
# ENCAPSULATION

- Encapsulation: Each layer adds headers/trailers around data.
- Process: Application Data → TCP/UDP Segment → IP Packet → Ethernet Frame.
- Transmission: bits on wire; Reception: decapsulation removes headers.
- Purpose: interoperability and abstraction between layers.



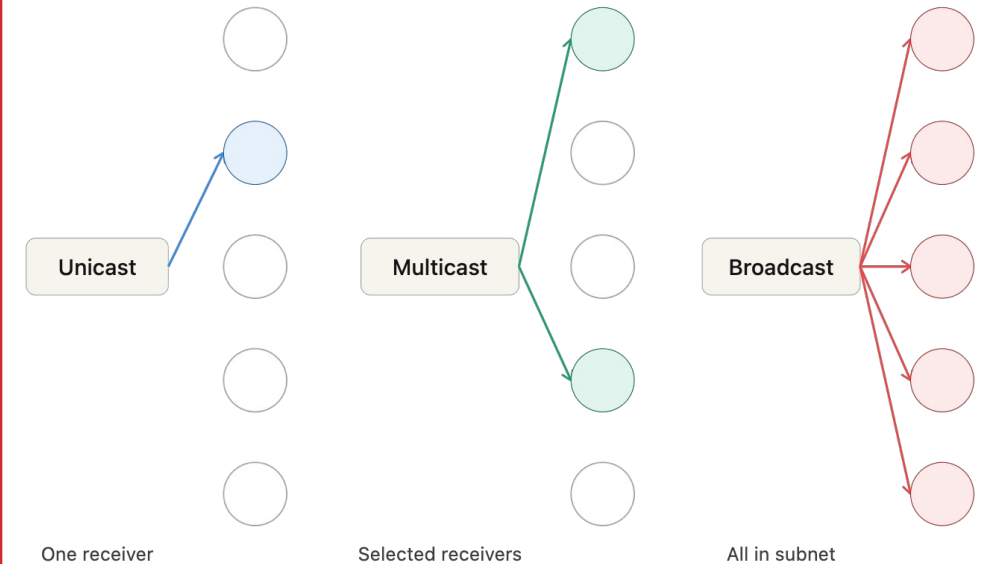
# MAC ADDRESSING & ARP OVERVIEW

- MAC Address: 48-bit hardware ID, unique per device (e.g., 00:1C:42:2B:60:5A).
- ARP: Resolves IP → MAC within a local network (broadcast request, unicast reply).
- ARP Cache: stores mappings; cleared periodically.
- IPv6 uses Neighbor Discovery instead of ARP.



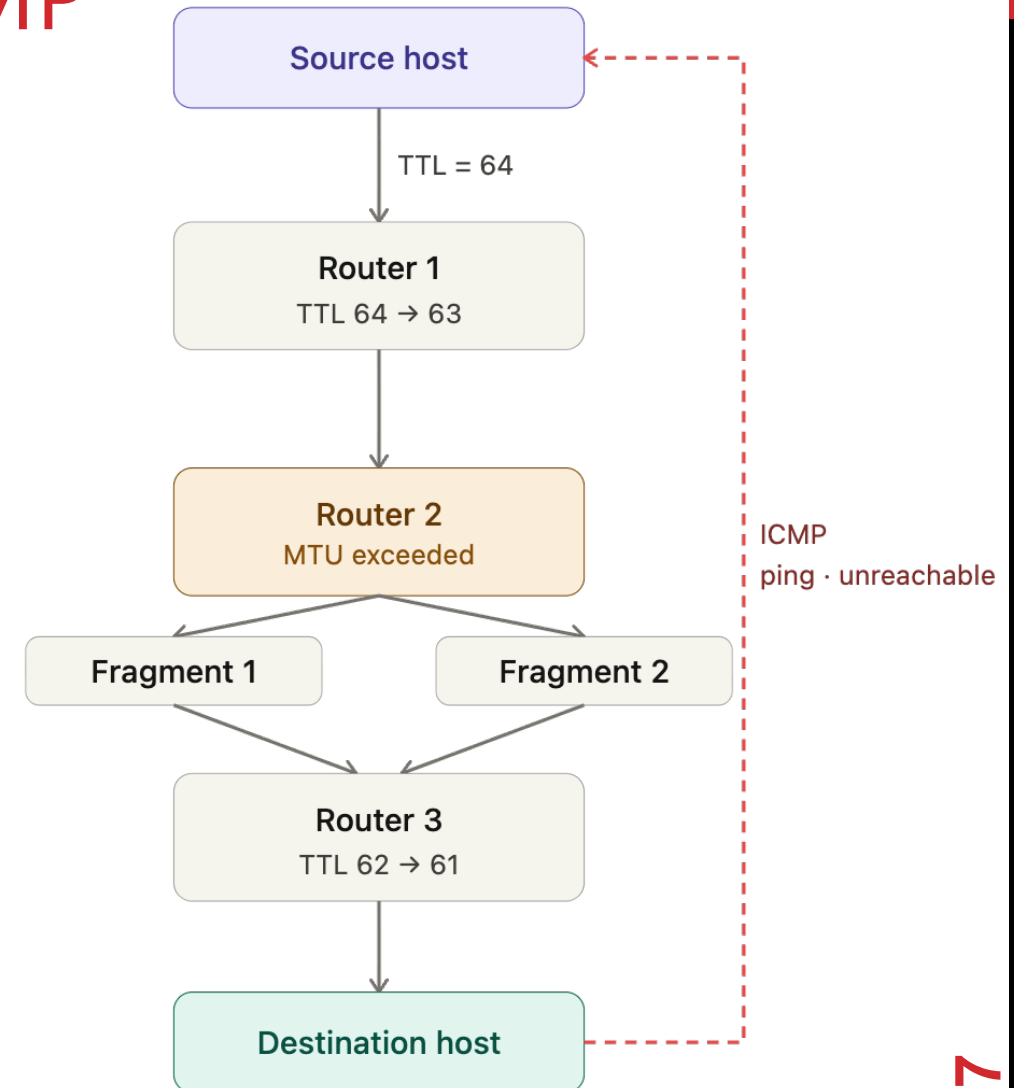
# UNICAST VS BROADCAST VS MULTICAST

- Unicast: one-to-one communication (most traffic).
- Broadcast: one-to-all in a subnet; routers block broadcasts.
- Multicast: one-to-many for groups; receivers opt-in (IGMP/MLD).
- IPv4 multicast 224.0.0.0/4;  
IPv6 multicast ff00::/8.



# MTU, FRAGMENTATION, TTL, ICMP

- MTU: Maximum Transmission Unit (e.g., Ethernet 1500 bytes).
- Fragmentation: IPv4 may fragment; IPv6 doesn't (uses Path MTU Discovery).
- TTL/Hop Limit: prevents loops, decremented per hop.
- ICMP: sends control messages (ping, unreachable, time exceeded).



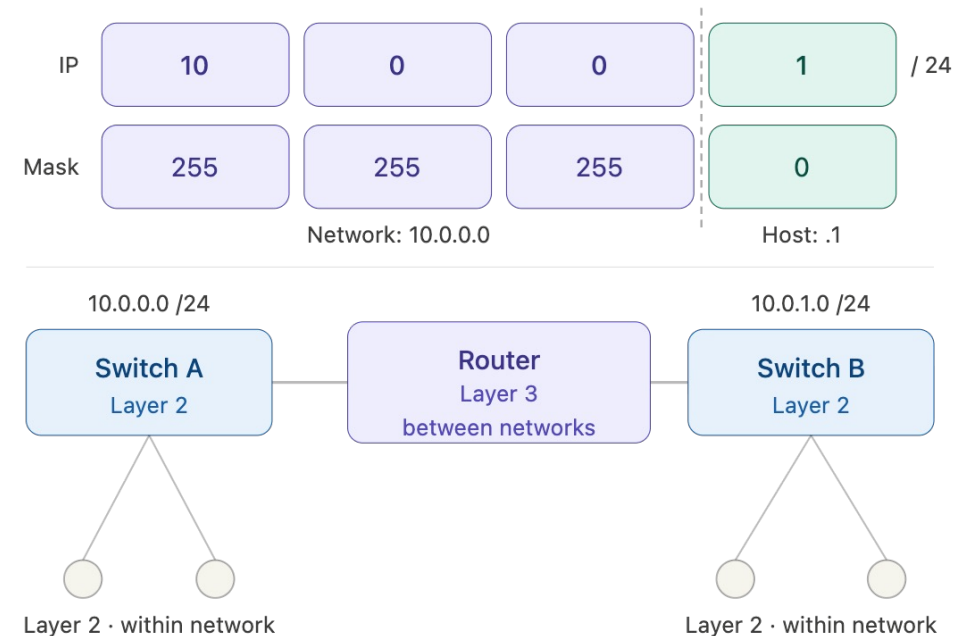


# PHYSICAL MEDIA: COPPER/FIBER, DUPLEX, AUTO-MDIX

- Copper: twisted pair (Cat5e–Cat7); 100m max; sensitive to EMI.
- Fiber: single-mode (long distance), multi-mode (shorter).
- Duplex: full (simultaneous) vs half; mismatches cause errors.
- Auto-MDIX: automatic cross-over correction on Ethernet ports.

# IP ADDRESSING 101: NETWORK VS HOST

- IP Address = Network Portion + Host Portion.
- Examples : 10.0.0.1/24
- Subnet mask defines boundary between network and host.
- 255.255.255.0 -> available ip addresses
- Routers forward between networks; switches forward within one network.



# PRIVATE VS PUBLIC RANGES (RFC1918)

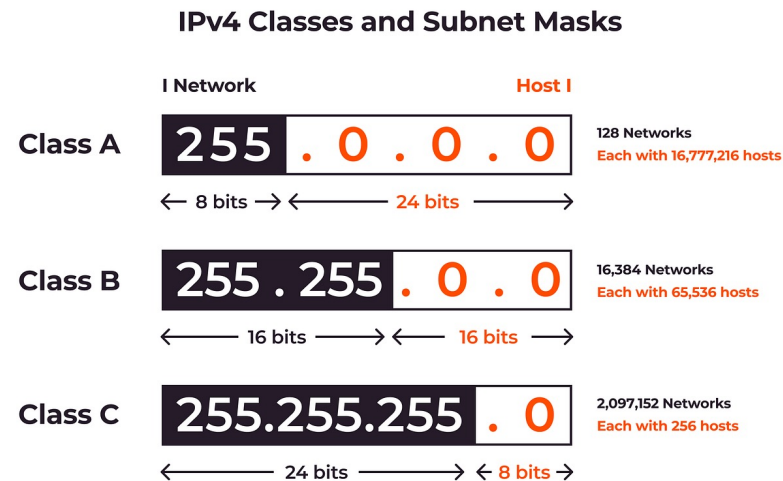
- Private IPv4 ranges: 10.0.0.0/8, 172.16.0.0–172.31.0.0/12, 192.168.0.0/16.
- Public IPs are routable on the Internet; private are not.
- NAT ( Network Address Translation ) translates private ↔ public addresses.
- RFC1918 defines private use ranges.

# CIDR NOTATION & SUBNET MASKS

- CIDR: Classless Inter-Domain Routing, written as /N (e.g., 192.168.1.0/24).
- Subnet mask determines prefix length: /24 = 255.255.255.0.
- CIDR enables efficient IP allocation and reduces routing table size.

# VLSM CONCEPT AND BENEFITS

- VLSM: Variable Length Subnet Masking – using different subnet sizes.
- Allows optimal address usage by assigning smaller/larger subnets as needed.
- Enhances flexibility and reduces waste of IP addresses.



# SUBNETTING METHOD

- Step 1: Determine required subnets and hosts.
- Step 2: **Borrow** bits from host portion for subnets.
- Step 3: **Calculate** subnet **ranges**, broadcast, and host counts.
- Formula:  $\text{Hosts} = 2^h - 2$ .



# SUBNETTING EXAMPLE 1 (/24 TO /26, /27)

- Start with 192.168.10.0/24 (256 addresses).
- /26 → 64 addresses per subnet (62 usable).
- /27 → 32 addresses per subnet (30 usable).
- Demonstrate creating four /26 subnets and eight /27 subnets.

| Subnet | Network           | First Usable   | Last Usable    | Broadcast      |
|--------|-------------------|----------------|----------------|----------------|
| 1      | 192.168.10.0/26   | 192.168.10.1   | 192.168.10.62  | 192.168.10.63  |
| 2      | 192.168.10.64/26  | 192.168.10.65  | 192.168.10.126 | 192.168.10.127 |
| 3      | 192.168.10.128/26 | 192.168.10.129 | 192.168.10.190 | 192.168.10.191 |
| 4      | 192.168.10.192/26 | 192.168.10.193 | 192.168.10.254 | 192.168.10.255 |

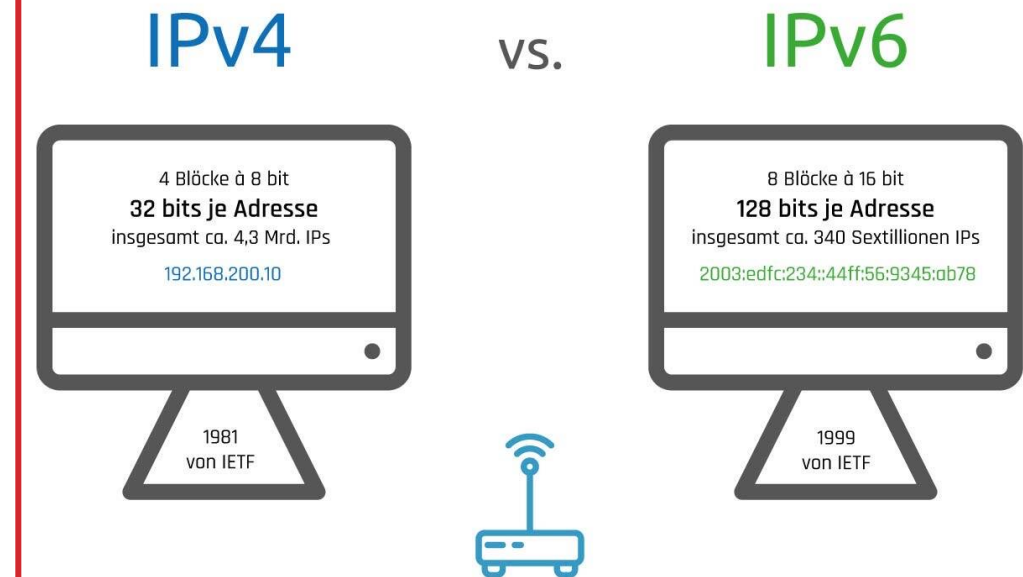
# SUBNETTING EXAMPLE 2 (VLSM FOR MIXED SIZES)

- Example: Need subnets for 50 hosts, 20 hosts, 10 hosts.
- Start from largest: allocate /26 (62 hosts), then /27 (30 hosts), then /28 (14 hosts).
- VLSM prevents waste of addresses.

| Requirement | Subnet | Network       | Usable Hosts      | Broadcast      |
|-------------|--------|---------------|-------------------|----------------|
| 50 hosts    | /26    | 192.168.10.0  | 192.168.10.1–62   | 192.168.10.63  |
| 20 hosts    | /27    | 192.168.10.64 | 192.168.10.65–94  | 192.168.10.95  |
| 10 hosts    | /28    | 192.168.10.96 | 192.168.10.97–110 | 192.168.10.111 |

# IPV6 BASICS: NOTATION, TYPES, SCOPE

- IPv6 = 128-bit address written in hexadecimal (e.g., 2001:db8::1).
- No broadcast; supports unicast, multicast, anycast.
- Abbreviations: omit leading zeros, use :: for consecutive zeros once.



# LINK-LOCAL VS GLOBAL UNICAST, EUI-64

- Link-local (fe80::/10): auto-configured, used on local link only.
- Global unicast: routable on Internet (e.g., 2000::/3).
- EUI-64: generates interface ID from MAC (inserts ff:fe and flips bit).

# TOOLS: PING, TRACEROUTE, IPCONFIG/IFCONFIG

- ping: tests reachability using ICMP Echo.
- traceroute/tracert: displays path and latency per hop.
- ipconfig (Windows) / ifconfig or ip (Linux): shows IP configuration.
- nslookup / dig: DNS resolution diagnostics.

# POP QUIZ:

What is the subnet mask for a /26 network?

- A. 255.255.255.0
- B. 255.255.255.192
- C. 255.255.255.224
- D. 255.255.255.240





# POP QUIZ:

What is the subnet mask for a /26 network?

- A. 255.255.255.0
- B. **255.255.255.192**
- C. 255.255.255.224
- D. 255.255.255.240





# POP QUIZ:

Which of the following is the broadcast address of the subnet 192.168.10.0/27?

- A. 192.168.10.31
- B. 192.168.10.32
- C. 192.168.10.30
- D. 192.168.10.1



# POP QUIZ:

Which of the following is the broadcast address of the subnet 192.168.10.0/27?

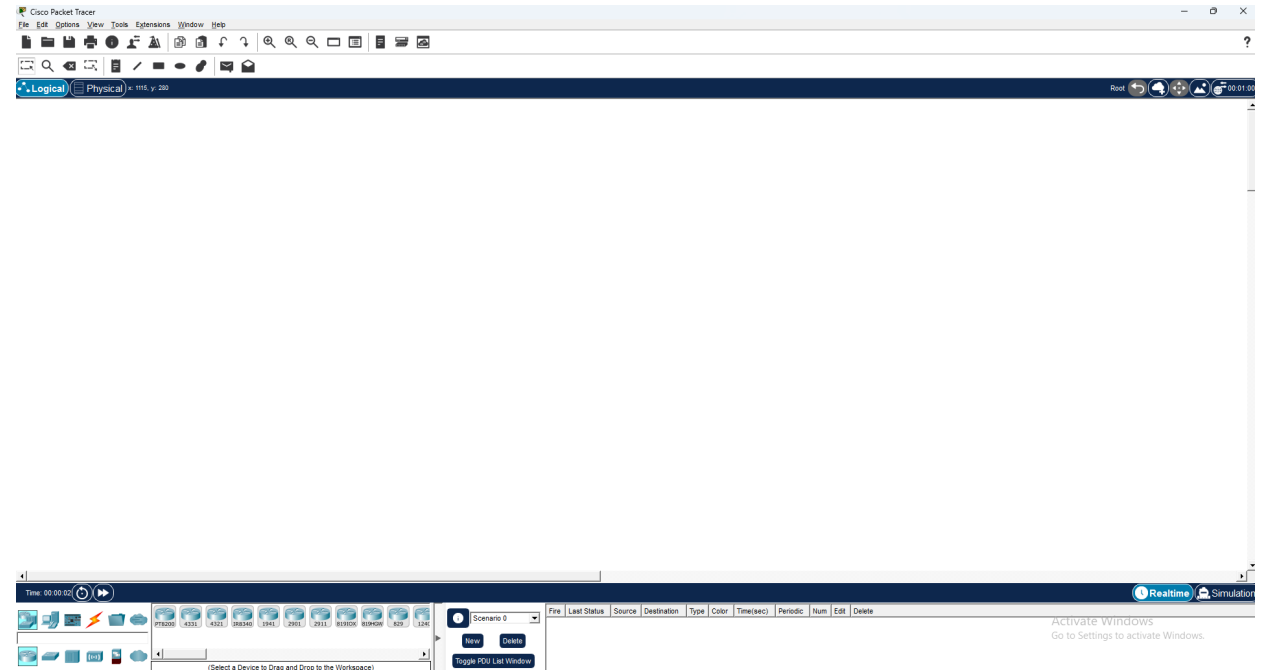
- A. **192.168.10.31**
- B. 192.168.10.32
- C. 192.168.10.30
- D. 192.168.10.1



# CLIS & CISCO PACKET TRACER

# INTRODUCTION TO CISCO PACKET TRACER

- Free Cisco simulation tool for network configuration and troubleshooting
- Visualizes packet flow between devices
- Supports routers, switches, PCs, servers, and IoT devices
- Ideal for CCNA-level labs and classroom demos



# KEY FEATURES AND SIMULATION MODES

- Free Cisco simulation tool for network configuration and troubleshooting
- Visualizes packet flow between devices
- Supports routers, switches, PCs, servers, and IoT devices
- Ideal for CCNA-level labs and classroom demos



# LAB DESIGN TIPS

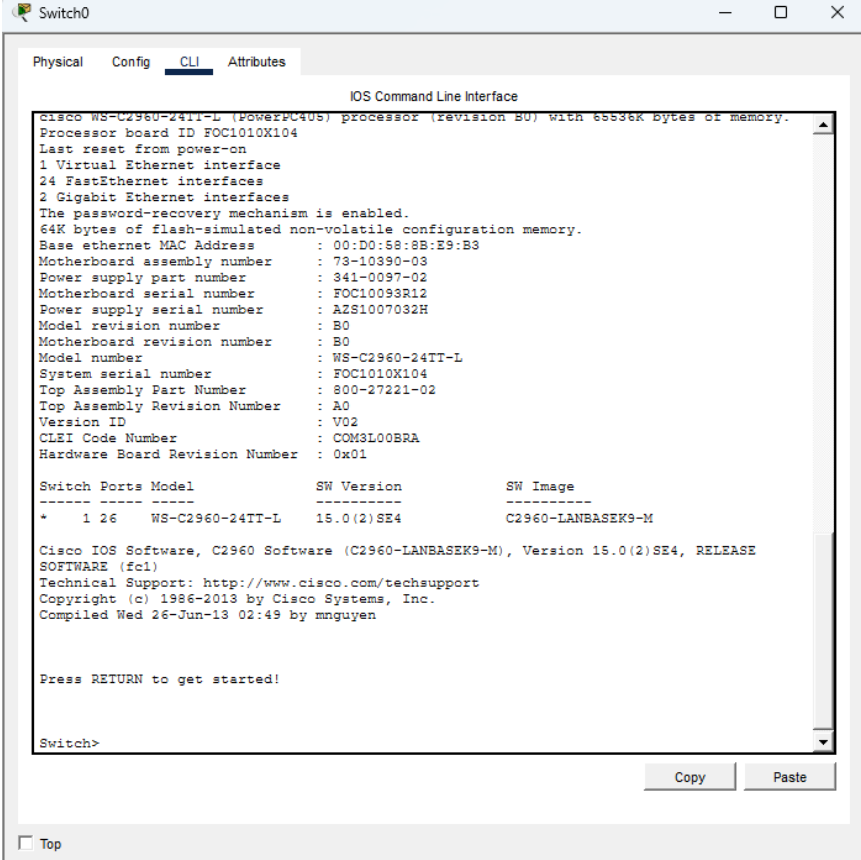
- Start small: test connectivity between 2–3 nodes
- Add routers and subnets progressively
- Use labels for IPs and interfaces
- Save snapshots for each step
- Verify using ping, show ip int brief, and arp





# CISCO IOS OVERVIEW

- Cisco IOS = Internetwork Operating System
- Runs on Cisco routers and switches
- Hierarchical CLI: User EXEC → Privileged EXEC → Global Config → Interface Config
- Uses context-sensitive help (?)



```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
cisco WS-C2960-24TT-L (PowerPC405) processor (revision B0) with 65536K bytes of memory.
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 00:D0:58:8B:E9:B3
Motherboard assembly number    : 73-10390-03
Power supply part number       : 341-0097-02
Motherboard serial number      : FOC10093R12
Power supply serial number     : AZS1007032H
Model revision number          : B0
Motherboard revision number    : B0
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010X104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A0
Version ID                     : V02
CLEI Code Number               : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version  SW Image
-----
* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

Switch>
```



# ESSENTIAL IOS COMMANDS

- show ip int brief – list interfaces
- show running-config / show startup-config
- configure terminal, interface g0/0, ip address ..., no shutdown
- copy running-config startup-config

# IOS TROUBLESHOOTING TOOLS

- ping and traceroute
- show arp, show mac address-table
- debug ip packet, show logging
- clear counters, reload



# ARISTA EOS OVERVIEW

- EOS = Extensible Operating System
- Based on Linux (Fedora Core)
- Modular architecture with independent processes (SysDB, agents)
- CLI compatible with Cisco IOS (similar syntax)

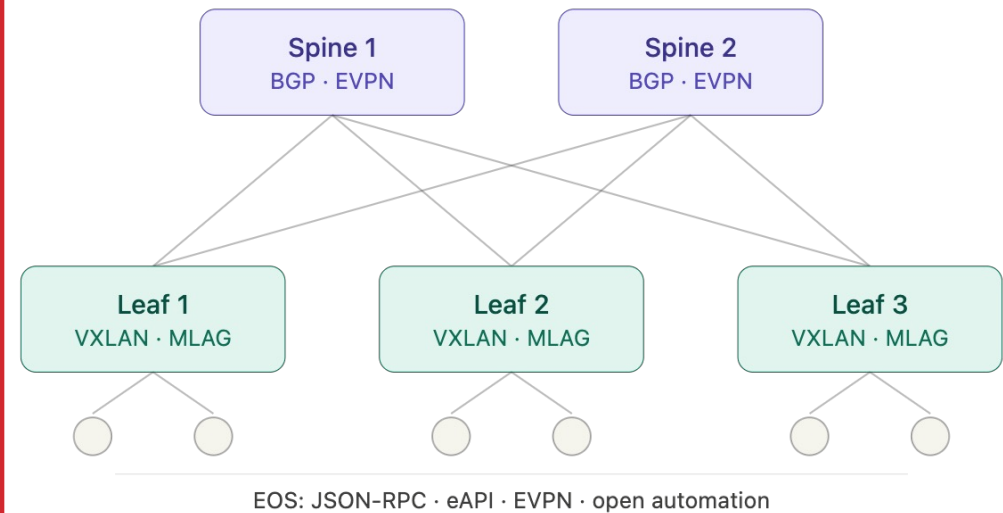


# ARISTA EOS CLI & CONFIGURATION

- Privilege levels like Cisco IOS (enable, configure terminal)
- Common commands:
  - show interfaces status
  - show ip route
  - copy running-config startup-config
- Supports eAPI, Python, and Ansible for automation

# EOS IN MODERN NETWORKS

- Used in spine-leaf architectures and data centers
- Supports advanced features: VXLAN, MLAG, EVPN
- Programmable and API-driven (JSON-RPC)
- Backed by a strong open automation ecosystem



# POP QUIZ:

Which of these commands runs on the ICMP protocol?

- A. ssh
- B. ping
- C. telnet
- D. curl



# POP QUIZ:

Which of these commands runs on the ICMP protocol?

- A. ssh
- B. ping**
- C. telnet
- D. curl





# POP QUIZ:

Which of the following is a basic Cisco IOS CLI command?

- A.** show ip interface brief
- B. apt-get install
- C. systemctl status
- D. kubectl get pods



# POP QUIZ:

Which of the following is a basic Cisco IOS CLI command?

A. **show ip interface brief**

B. apt-get install

C. systemctl status

D. kubectl get pods



# LAB 00: PACKET TRACER DISCOVERY

- Goal: Discover CISCO Packet Tracer
- Steps:
  - Navigate through panels
  - Drag and drop network devices and endpoints
  - Connect components

# LAB 01: NETWORKING FUNDAMENTALS WITH CISCO

- Goal: Master basic networking concepts and packet analysis.
- Steps:
  - Build basic LAN topology with PCs and switch.
  - Configure IP addressing and test connectivity.
  - Analyze ARP tables and MAC address learning.
  - Observe packet encapsulation in simulation mode.

# LAB 02: VLSM ADDRESS DESIGN

- Goal: Design efficient IP addressing using VLSM.
- Steps:
  - Calculate subnet requirements for different departments.
  - Apply VLSM to minimize IP waste.
  - Configure subnets on router interfaces.
  - Verify routing between subnets.

# LAB 03: IPV6 CONFIGURATION

- Goal: Implement IPv6 addressing and routing.
- Steps:
  - Configure IPv6 addresses on devices.
  - Enable IPv6 routing on router.
  - Test IPv6 connectivity with ping6.
  - Compare IPv4 and IPv6 address structures.